

Isaac Mubiru

Senior Software Engineer — Backend & Cloud Systems

isaacmubiru99@gmail.com | +256 702 360 601 | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 |

Kampala, Uganda (EAT, UTC+3)

PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years building and scaling production backend systems for remote-first companies in the US, Singapore, and East Africa. Led the end-to-end migration of 20+ applications from Parse/MongoDB to Firebase and Google Cloud Platform at MTailor with zero downtime — building a real-time two-way sync pipeline so live traffic ran uninterrupted throughout — and cut infrastructure costs by \$5,000/month. Deep expertise in Node.js, Firebase, and GCP, with hands-on experience across microservices (Kafka, Docker, Kubernetes), real-time data pipelines (Pub/Sub), and full-stack delivery from React frontends to USSD services. Comfortable owning systems end to end: architecture, implementation, deployment, documentation, and team leadership.

CORE COMPETENCIES

Node.js Backend Systems

Zero-Downtime Cloud Migrations

Firebase & Google Cloud Platform

Real-Time Data Pipelines (Pub/Sub)

Microservices (Kafka, Kubernetes)

API Design & Integrations

React & Cross-Platform Frontends

Technical Team Leadership

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub, allowing legacy and new systems to serve production traffic in parallel during the migration.
- Saved the company **\$5,000/month** by migrating all services off AWS.
- Built a 3D visualisation feature for customers using video overlay and ffmpeg, increasing buyer conversion.
- Implemented Express Shipping, generating an additional **\$40 revenue per order**.
- Migrated file storage from Amazon S3 to Google Cloud Storage via a Python script, and moved a Webflow website from a Parse backend to Firebase.

Node.js · Firebase · GCP Pub/Sub · AWS (S3, EC2, EBS) · Python · ffmpeg · React · Webflow · JavaScript

MTailor Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore and GCP; performance led to a full-time offer.
- Built a migration script to read data from MongoDB and write to Firestore after running complex processing logic.
- Wrote documentation for fellow engineers on working with the new Firebase SDKs (Node.js and Python).
- Trained the Ops team on the new Firebase dashboard and backup procedures.

Node.js · Firebase · AWS (S3, EC2, EBS) · Google Cloud Platform · JavaScript

Mind2Matter Jan 2022 – Jul 2022

Software Engineer · US Remote

- Full-stack development for a US-based agency with demanding clients, reporting directly to the CTO.
- Built backends for DeFi applications using Web3 and Node.js.
- Delivered React UIs from Figma designs under tight timelines.

Node.js · Firebase · React · Web3 · JavaScript · Tailwind · HTML · CSS

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices from a Firebase backend.
- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer prices current from Morningstar APIs.
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku.
- Added responsive UI pages to the PWA using HTML, Tailwind, and React.

Node.js · Firebase · React · JavaScript · Tailwind · HTML · CSS

CodeBits

Jan 2020 – Jul 2021

Software Engineer & Team Lead · Uganda Remote

- Led a team of **4 developers** building legal tech systems for NGOs in Uganda.
- Architected a microservices backend for FIDA Uganda's case management app using Apache Kafka, Docker, and Kubernetes.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.
- Developed a cross-platform mobile app for LASPNET using React Native and Expo (iOS & Android), with Firebase push notifications.
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC.

Node.js · Firebase · React Native · Apache Kafka · Docker · Kubernetes · NATS Streaming · gRPC · USSD · JavaScript

PROJECTS

Healthcare WhatsApp Chatbot

Built a healthcare assistant on the WhatsApp Business API, integrating OpenAI for conversational flows and Google Vision for prescription OCR — letting patients submit prescriptions as photos and get structured responses.

Node.js · WhatsApp Business API · OpenAI · Google Vision

Marketplace Recommendation Engine

Designed and built a recommendation engine combining collaborative and content-based filtering in MongoDB to match customers with service providers on a marketplace platform.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — [Makerere University](#)

2018 – 2022

SKILLS

Proficient: Node.js, Firebase, React, Google Cloud Platform, Python, MongoDB, SQL, HTML, CSS, Git, Linux

Intermediate: Go, AWS, Rust, React Native

Tools & Infrastructure: Docker, Kubernetes, Apache Kafka, Google Pub/Sub, gRPC, Express.js, Playwright, ffmpeg

Languages: English (fluent), Kiswahili